

Swaham Pitambar Patra

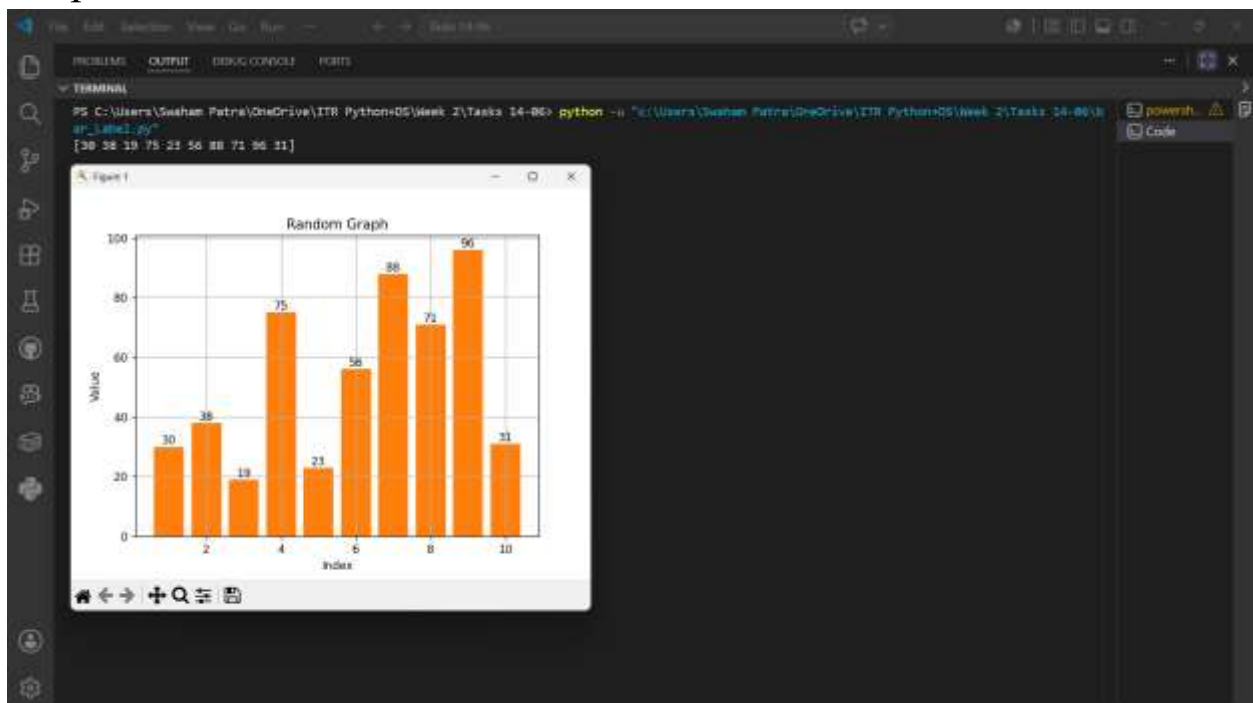
Tasks:

Bar label:

Code:

```
bar_label.py X
bar_label.py > ...
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 data = np.random.randint(1,100,10)
5 print(data)
6
7 bar = plt.bar(range(1,11), data)
8
9 plt.bar_label(bar)
10
11 plt.bar(range(1,11), data)
12 plt.title("Random Graph")
13 plt.xlabel("Index")
14 plt.ylabel("Value")
15 plt.grid()
16
17 plt.show()
```

Output:

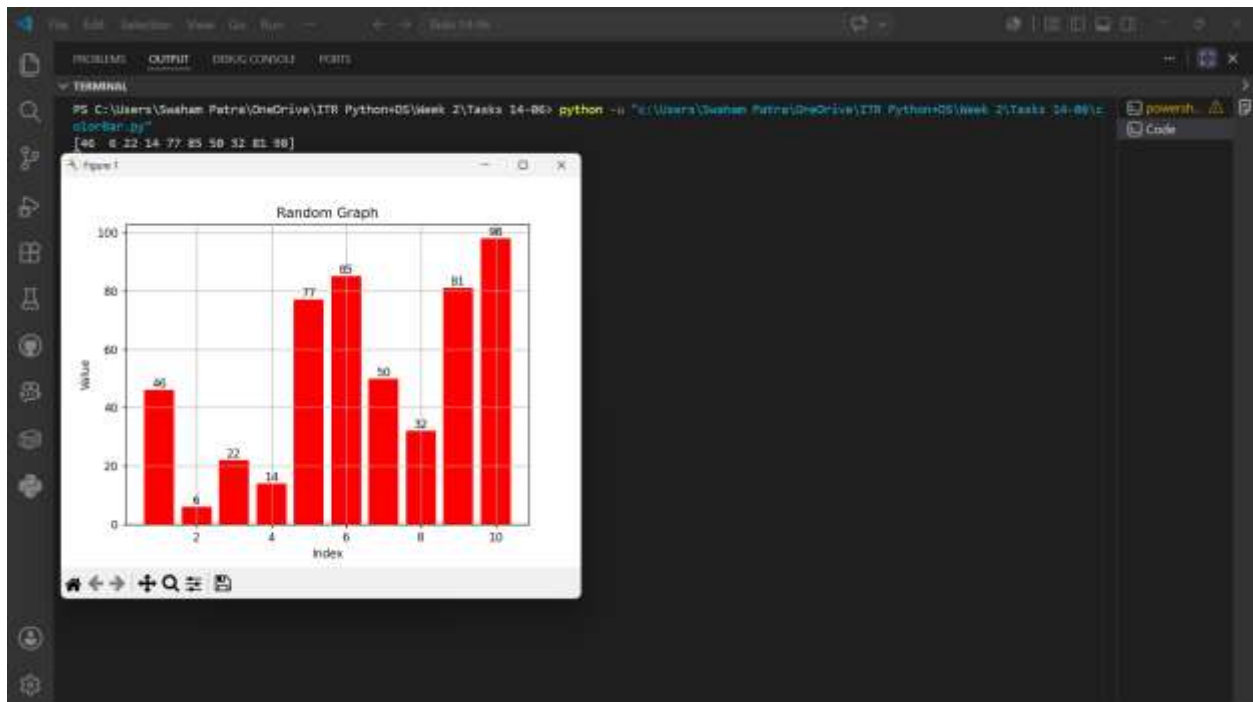


Change colour of graph:

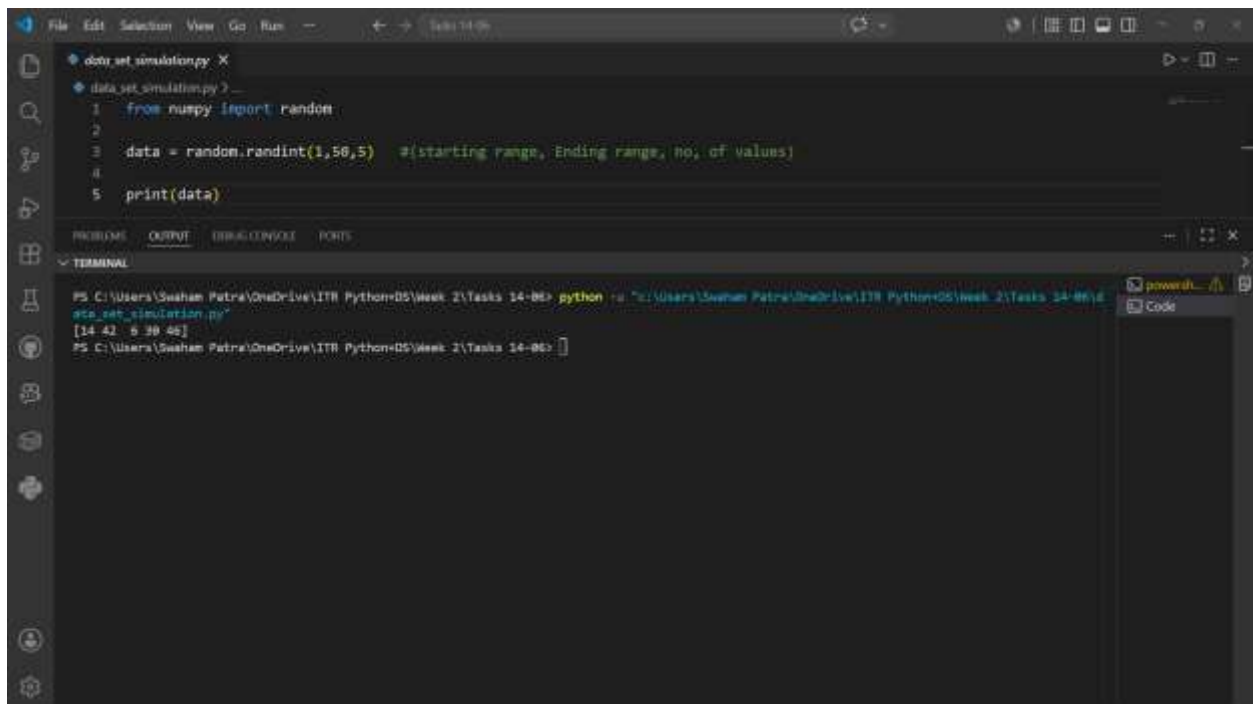
Code:

```
colordbar.py X
colordbar.py -
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 data = np.random.randint(1,100,10)
5 print(data)
6
7 bar = plt.bar(range(1,11), data)
8
9 plt.bar_label(bar)
10
11 plt.bar(range(1,11), data ,color="red")
12 plt.title("Random Graph")
13 plt.xlabel("Index")
14 plt.ylabel("Value")
15 plt.grid()
16
17 plt.show()
```

Output:



Data set simulation:



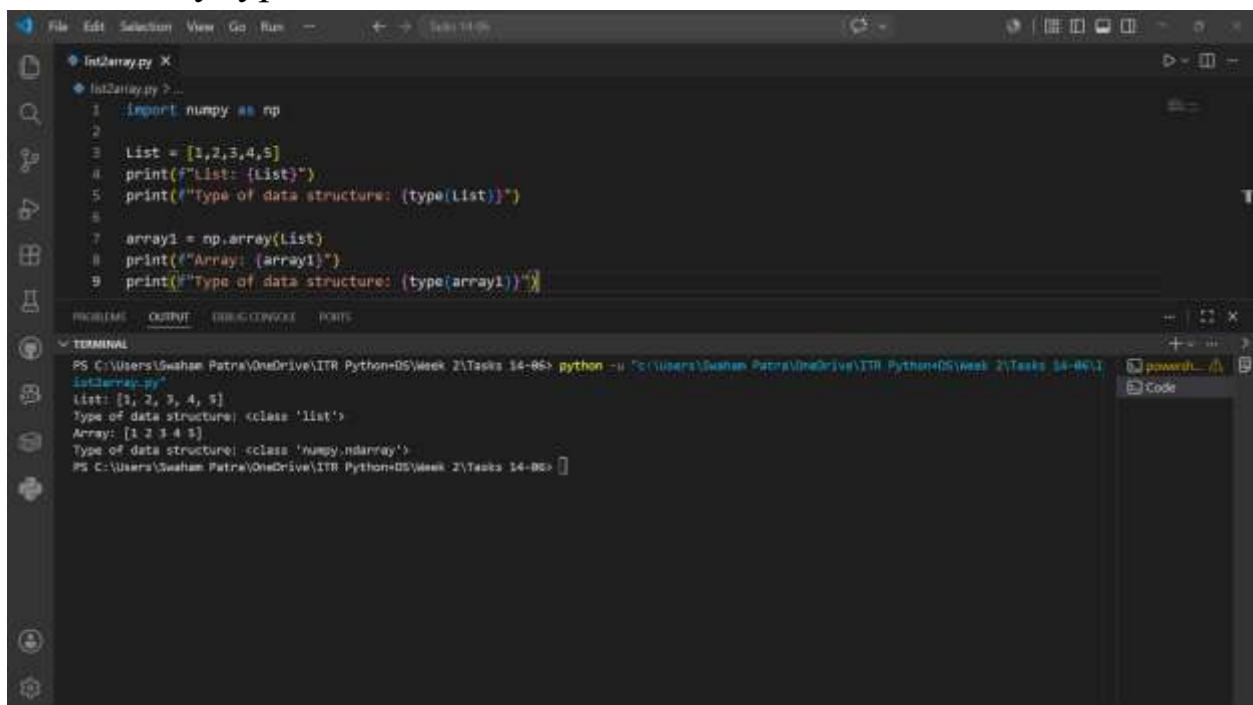
```
File Edit Selection View Go Run - Task 14-86
data_set_simulation.py X
data_set_simulation.py > ...
1 from numpy import random
2
3 data = random.randint(1,50,5) #(starting range, Ending range, no. of values)
4
5 print(data)
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS\Week 2\Tasks 14-86> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\DS\Week 2\Tasks 14-86\data_set_simulation.py"
[14 42  5 39 46]
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS\Week 2\Tasks 14-86>
```

List to array type cast:



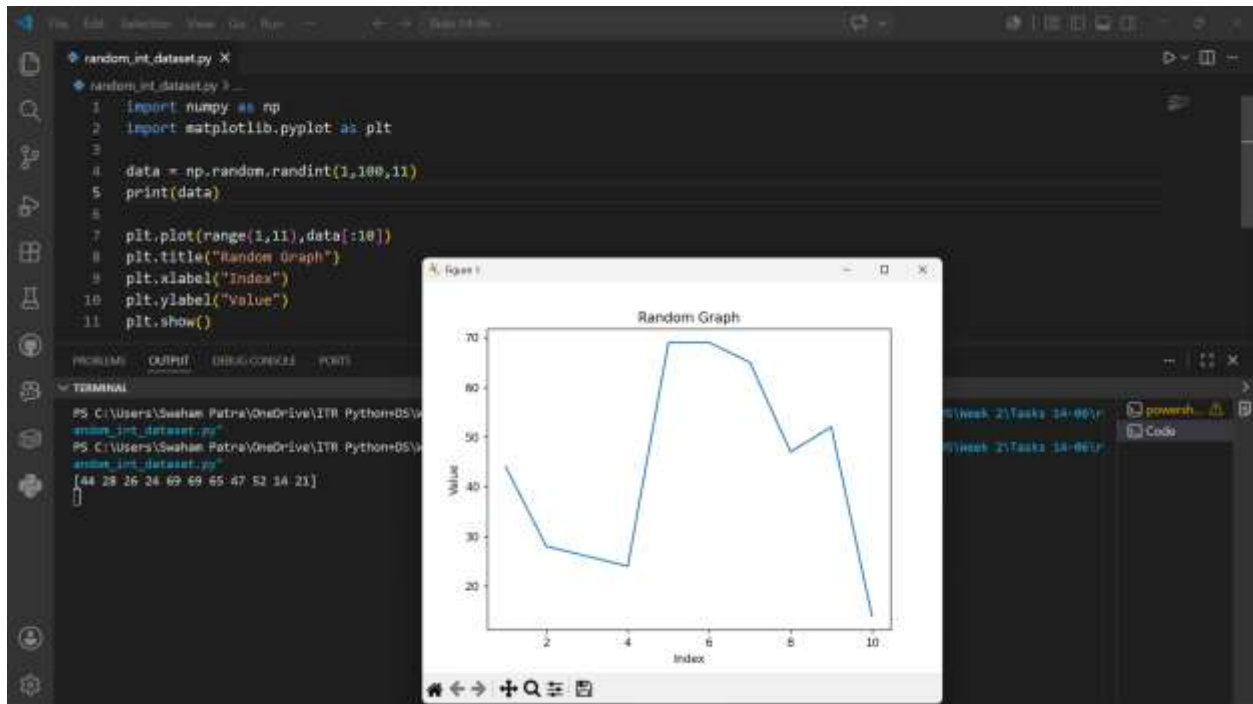
```
File Edit Selection View Go Run - Task 14-86
list2array.py X
list2array.py > ...
1 import numpy as np
2
3 List = [1,2,3,4,5]
4 print(f"List: {List}")
5 print(f"Type of data structure: {type(List)}")
6
7 array1 = np.array(List)
8 print(f"Array: {array1}")
9 print(f"Type of data structure: {type(array1)}")
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS\Week 2\Tasks 14-86> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\DS\Week 2\Tasks 14-86\list2array.py"
list: [1, 2, 3, 4, 5]
Type of data structure: <class 'list'>
Array: [1 2 3 4 5]
Type of data structure: <class 'numpy.ndarray'>
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS\Week 2\Tasks 14-86>
```

Use of random in numpy:



Use of vector in python:

